SOIL EROSION AND CONSERVATION · NOTE ST06

Earth Dams and Flood Routing — Elements, Types, Design Criteria, Filters and Reservoir Routing

Course topics: SWCE-503 T3, SWCE-503 T7, SWCE8.12 L9, SWCE8.12 L10, SWCE8.12 L5, SWCE3.5 L14

Earth dams

Built to store water for irrigation, domestic use, livestock and recreation, earth dams are the commonest small dams because their materials are used almost as found. The design must suit the foundation and the available fill.

Element	Description
Shell (upstream/downstream fill, shoulder)	Pervious or semi-pervious fill — upstream and downstream shells
Upstream blanket	Impervious layer on the upstream side over a pervious substratum: cuts seepage, lengthens the flow path and lowers seepage pressure downstream (a natural blanket is a low-permeability soil cover)
Drainage filter	Blanket of pervious material from the foundation to the downstream side that discharges seepage and prevents piping
Cutoff wall	Structure in or on the foundation that reduces percolation through porous strata
Riprap	Broken stone on the slopes against waves and erosion
Core wall (membrane)	Central impervious wall of puddled clay, masonry or concrete that checks flow through the dam
Toe drain	Drain at the downstream slope that collects and removes seepage
Transition filter	Intermediate material between core and shells — filters and stops fines moving sideways
Cutoff trench	Excavation at the base filled with impervious material to reduce seepage
Key wall	Cutoff wall carried into bedrock or an impervious stratum
Inverted (protective) filter	Layers of pervious or semi-pervious material that aid drainage and stop soil moving
Other parts	Impervious upstream fill, pervious soil bed, grass cover on the downstream slope, downstream toe, semi-pervious downstream fill

- **Line of seepage (phreatic surface):** the surface — joining points at atmospheric pressure — that separates saturated from unsaturated soil. Stored water gives a pressure head equal to its depth, falling toward the downstream side; the slope of the saturation line is the saturation gradient. The line must meet the base **within** the dam section.
- **Filter:** graded coarse material, usually gravel, at the downstream base; it keeps the saturation line off the toe and draws seepage into itself — a rock-fill toe drain on an impervious foundation, or graded

gravel enveloping rock fill on a pervious one.

- **Breaching:** overtopping destroys the dam and damages the area below — provide emergency spillways of adequate capacity and protect inlet, conduit and outlet so water leaves the reservoir safely.

Foundation and fill

- Determine soil properties by laboratory and field tests. The best foundation is a thick impervious layer: remove topsoil and scarify. Bedrock is usually safe, apart from cracks and fissures.
- **Pervious foundation:** a cutoff trench parallel to the dam centre line, reaching the impervious layer — bottom width at least 1.2 m, side slopes 1 : 1 — backfilled in layers of impervious soil compacted at optimum moisture.
- **Fill:** impervious and stable (sandy soil is unsuitable), free of vegetation, grass, leaves and twigs. If materials vary, put impervious soil in the upstream 2/3 and pervious soil in the downstream 1/3. Place in 20 cm layers and compact at optimum moisture with mechanical or sheep-foot rollers, bonding each layer to the last.

Type	Features
Homogeneous	Very small dams of one impervious material (besides riprap); fairly flat slopes upstream and downstream to prevent sloughing — sliding of saturated overlying material
Zoned (most common)	Several materials: impervious core, semi-pervious zone round it, pervious sand and gravel shells, and a drainage system (horizontal drain or rock toe)
Diaphragm	A modified homogeneous dam: pervious embankment with a thin central impervious diaphragm of compacted clay from the foundation to near the top

Correction: the notes say the diaphragm "must not be less than 10 m" thick. The standard rule is the opposite — a diaphragm is thinner than 10 m and thinner than the embankment height at that level; if it is thicker than 10 m, the dam is classed as zoned (as the notes themselves go on to say).

Design criteria

$$\text{Top width, low dams: } T_w = H/5 + 3$$

$$\text{Top width, medium dams } (H < 30 \text{ m}): T_w = 0.55\sqrt{H} + 0.2H \text{ } (T_w, H \text{ in m; at least 4.5 m with a road on top})$$

$$\text{Storage between two contours: } V = \text{vertical interval} \times (\text{sum of the two contour areas})/2$$

$$\text{Filter: } D_{15}(\text{filter}) \leq 4 \text{ to } 5 \times D_{85}(\text{base}) \text{ and } D_{15}(\text{filter}) \geq 4 \text{ to } 5 \times D_{15}(\text{base})$$

- **Top width** must keep the phreatic line safely inside the section.
- **Freeboard** (top of embankment to high flood level after settlement): 2 to 3 m; the notes add a design criterion of "≈ 15" without a unit. It protects against overtopping, waves and excess runoff.
- **Settlement** from consolidation and wetting: increase the dam height by 5%.
- **Side slopes:** upstream 4 : 1, downstream 3 : 1 — flatter is better, given water-level changes, waves and sloughing.
- **Failure** comes from seepage, piping (seepage water passing through the body or foundation — prevented by proper design and filters) and sloughing.

Illustration (not in the notes)
Low dam H = 6 m → T_w = 6/5 + 3 = 4.2 m. Medium dam H = 16 m → T_w = 0.55 × 4 + 0.2 × 16 = 5.4 m

Flood routing

Storage–discharge relations of flood-control reservoirs are designed by flood routing: finding inflow, outflow and storage at set time intervals. Evaporation and seepage are neglected over a flood.

Continuity: $i \cdot dt = o \cdot dt + s \cdot dt$ (i, o, s = inflow, outflow and storage rates, L^3/T)
Routing equation: $\Delta t \cdot (I_1 + I_2)/2 = \Delta t \cdot (O_1 + O_2)/2 + S_2 - S_1$ (1 = start, 2 = end of interval; S = storage volume)

- **Four basic elements:** the inflow hydrograph (the runoff hydrograph); the spillway discharge curve (depth vs discharge, main plus emergency spillway); the available-storage curve (depth vs storage above the spillway crest, from basin topography by the average-end-area method); and the outflow hydrograph, which routing produces.
- **Methods:** (1) the **time-interval method** — split the inflow hydrograph into short intervals with inflow and outflow taken as constant in each, using the discharge and storage curves; (2) the **analytical method** — integrate, replacing the flood by an equivalent uniform discharge. In practice a partly numerical, partly graphical method is used.
- **Spillway size by trial:** assume a size, route the flood, find the peak outflow and maximum stage, and repeat with a new size if unsatisfactory. For drop-inlet pipe spillways, the discharge when the pipe first flows full is slightly less than the maximum outflow.

Culp (1948), drop-inlet pipe spillway: $q_0/q = 1.25 - (1500V/(R \cdot A) + 0.06)\text{^}(1/2)$
q_0 = outflow when the pipe first flows full (m^3/s); q = peak inflow (m^3/s); V = available storage (ha-m); R = runoff (mm); A = drainage area (ha)

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. US Bureau of Reclamation, Design of Small Dams and Hydraulic Design of Stilling Basins and Energy Dissipators (EM 25). S. K. Garg, Irrigation Engineering and Hydraulic Structures (earth-dam criteria). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. The most common type of earth dam is
(a) homogeneous
(b) zoned
(c) diaphragm
(d) rock-fill

Q2. Top width of a low earth dam is
(a) H/5 + 3
(b) 0.55√H + 0.2H
(c) H/2
(d) 3H

Q3. Minimum bottom width of a cutoff trench is
(a) 0.6 m
(b) 1.2 m
(c) 3 m
(d) 5 m

Q4. Earth fill is placed in layers of about
(a) 5 cm
(b) 20 cm
(c) 50 cm
(d) 1 m

Q5. Settlement allowance in dam height is

(a) 1%
(b) 5%
(c) 15%
(d) 25%

Q6. In a dam of mixed materials, impervious soil goes in the

(a) downstream 1/3
(b) upstream 2/3
(c) centre 1/10
(d) toe

Q7. Filter stability criterion:

(a) $D_{15}f \le 4–5\ D_{85}b$
(b) $D_{85}f \le D_{15}b$
(c) $D_{50}f = D_{50}b$
(d) $D_{15}f \ge 10\ D_{85}b$

Q8. The flood-routing continuity equation is

(a) I = O
(b) $\Delta t(I_1 + I_2)/2 = \Delta t(O_1 + O_2)/2 + S_2 - S_1$
(c) S = I·t
(d) O = CLH^1.5

Q9. A diaphragm-type earth dam has a diaphragm thinner than

(a) 1 m
(b) 10 m and the embankment height
(c) 30 m
(d) 100 m

Answers

Q1 — (b) zoned
Q2 — (a) H/5 + 3
Q3 — (b) 1.2 m
Q4 — (b) 20 cm
Q5 — (b) 5%
Q6 — (b) upstream 2/3
Q7 — (a) $D_{15}f \le 4–5\ D_{85}b$
Q8 — (b) $\Delta t(I_1 + I_2)/2 = \Delta t(O_1 + O_2)/2 + S_2 - S_1$
Q9 — (b) 10 m and the embankment height